The limit behaviour of some phylogenetic tree balance indices

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Conference Discrete Random Structures II
Będlewo

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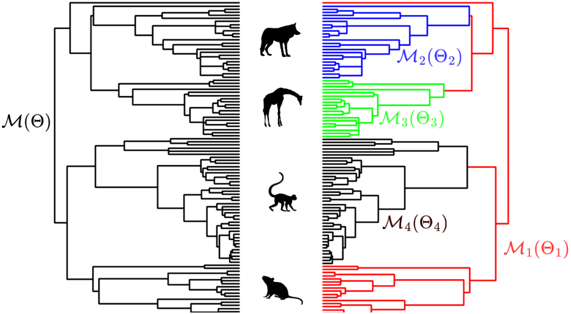
Phylogenetics

Joint work with the Computational Biology and Bioinformatics Research Group, Department of Mathematics and Computer Science, University of the Balearic Islands

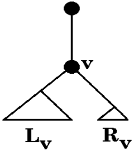


Graphic: Bayu Brahmantio

Aldous's β–model (tree on n leaves)

- q_n probability distribution in $\{1, \ldots, n-1\}$, s.t. $q_n(i) = q_n(n-i)$
- internal node v (n leaves) defines a split into two subtrees ($L_v, R_v = n - L_v$ leaves)

$L_v \sim \mathrm{BetaBinomial}(n, \beta+1, \beta+1)$, $-2 \leq \beta \leq \infty$



$$\begin{aligned} q_n(i) &= \tfrac{1}{a_n(\beta)} \tfrac{\Gamma(\beta+i+1)\Gamma(\beta+n-i+1)}{\Gamma(i+1)\Gamma(n-i+1)} \\ &= B(\beta+1, \beta+1)^{-1} \int_0^1 \left(\binom{n}{i}\tau^i(1-\tau)^{n-i}\right)\tau^\beta(1-\tau)^\beta \mathrm{d}\tau \end{aligned}$$

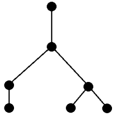
Special values

- $\beta = -3/2$ uniform model,
  all trees with same number of leaves have same probability
- $\beta = 0$ Yule tree,
  pure–birth branching process
- $\beta = \infty$ "random partition tree" model,

$$q_n(i) = \binom{n-2}{i-1} 2^{-(n-2)}$$

Balance indices

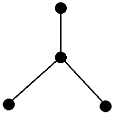
Colless': (binary) sum of balances (absolute difference between number of leaf descendents in left and right node) for each internal node



$$\sum_{v \text{ internal}} |L_v - R_v|$$

Sackin's: sum of distances from root of leaves of tree

Cophenetic:



$$\sum_{i,j \text{ leaf}} \phi_{ij},\ \phi_{ij} : \text{distance from root to mrca of } i,\ j$$

Quartet Index: number of B_4 quartets in tree

Contraction type distributions ($\tau \in [0, 1]$)

$\mathcal{D}$: space of mean–0, finite variance distributions
$S : \mathcal{D} \to \mathcal{D},\ S(F) = \mathcal{L}(\tau^r Y' + (1-\tau)^r Y'' + C(\tau)),\ Y', Y'' \sim F$

$$\tau \text{ s.t. } 2\sum_{i=1}^{n} p_{n,i}\left(\frac{i}{n}\right)^{2r} < 1, \quad p_{n,i} = P(((i-1)/n) < \tau \leq i/n)\ (*)$$

$$C(\tau) = \sum_{r_1+r_2\leq r} C_{r_1,r_2}\tau^{r_1}(1-\tau)^{r_2}$$

Rössler 1991 (A limit theorem for "Quicksort"):
S has a unique fixed point
S^n converges exponentially fast in Wasserstein d–metric

Wasserstein d–metric

$$d(F,G) = \inf \|X - Y\|_2$$

- $\|\cdot\|_2$ denotes the L_2 norm
- the infimum is over all pairs of random variables (X, Y) with marginals $X \sim F$, $Y \sim G$
- convergence in d induces convergence in distribution

Theorem: Contraction type limit $(n \geq 2, \beta > 0)$

$\{1,\ldots,n-1\} \ni L_n \sim \mathrm{BetaBinomial}(n-2,\beta+1,\beta+1)+1$
$\tau \sim \mathrm{Beta}(\beta+1,\beta+1)$

$$\begin{array}{rcl} Y_n & = & \left(\frac{L_n}{n}\right)^r Y_{L_n} + \left(1-\frac{L_n}{n}\right)^r Y_{n-L_n} + C_n(L_n) \\ C_n(i) & = & n^{-r}\left(\sum\limits_{r_1+r_2+r_3\leq r} C_{r_1,r_2,r_3} i^{r_1}(n-i)^{r_2} n^{r_3} + h_n(i)\right) \end{array}$$

$\mathrm{E}\left[C_n(L_n)\right]=0,\ \mathbb{N}_+ \ni r$ satisfies $(*)$, $\sup_i n^{-r} h_n(i) \to 0$
$\mathrm{E}\left[Y_n\right]=0,\ \mathrm{E}\left[Y_n^2\right]$ is uniformly bounded then,

$$Y_n \xrightarrow{W} Y_\infty \sim F \text{ s.t. } S(F)=F$$

Theorem: Contraction type limit (the challenge)

Yule tree ($\beta = 0$):

$L_n \sim \mathrm{Unif}\{1, \ldots, n-1\}$

$\tau \sim \mathrm{Unif}[0,1]$, $p_{n,i} = \frac{1}{n}$

$$n^{-1} L_n \stackrel{D}{=} n^{-1}(\lfloor (n-1)\tau \rfloor + 1)$$

$\beta > 0$: $L_n \sim \mathrm{BetaBinomial}(n-2, \beta+1, \beta+1) + 1$

$\tau \sim \mathrm{Beta}(\beta+1, \beta+1)$

$$p_{n,i} = B(\beta+1, \beta+1)^{-1} \int\limits_{(i-1)/n}^{i/n} u^{\beta}(1-u)^{\beta} \mathrm{d}u$$

$$n^{-1} L_n \stackrel{D}{\neq} n^{-1}(\lfloor (n-1)\tau \rfloor + 1)$$

QI Binary tree limit ($\beta \geq 0$)

$$QIB(T_n) = QIB(T_{L_n}) + QIB(T_{n-L_n}) + \binom{L_n}{2}\binom{n-L_n}{2}$$

$$Y_n^Q = n^{-4}\left(QIB(T_n) - \frac{3\beta+6}{7\beta+18}\binom{n}{4}\right)$$

For $\beta \geq 0$ and $\tau \sim \mathrm{Beta}(\beta+1, \beta+1)$ we have

$$Y_n^Q \xrightarrow{W} Y_Q$$

s.t. Y_Q satisfies

$$\begin{aligned} Y_Q \overset{D}{=} & \tau^4 Y_Q' + (1-\tau)^4 Y_Q'' + \frac{3\beta+6}{24(7\beta+18)}\left(\tau^4 + (1-\tau)^4\right) \\ & + \frac{1}{4}\tau^2(1-\tau)^2 - \frac{3\beta+6}{24(7\beta+18)} \end{aligned}$$

Simulation algorithm

procedure YRECURSION(m , Y_0)
    if $m = 0$ **then** $Y_1 = Y_0$, $Y_2 = Y_0$
    else
        $Y_1 =$YRECURSION($m - 1$, Y_0)
        $Y_2 =$YRECURSION($m - 1$, Y_0)
    end if
    Draw τ from F_τ
    return $\tau^r Y_1 + (1-\tau)^r Y_2 + C(\tau)$,
end procedure
return YRECURSION(M, Y_0)

Other Indices (Yule, $\beta = 0$)

Colless' (sum of internal node balances)

$$Y \overset{D}{=} \tau Y' + (1-\tau)Y'' + \tau \log \tau + (1-\tau)\log(1-\tau)$$
$$+1 - 2\min(\tau, 1-\tau)$$

Quadratic Colless' (sum of squares of internal node balances) $Y \overset{D}{=} \tau Y' + (1-\tau)Y'' + (1 + 6\tau + 6\tau^2)$

Sackin's (sum of node depths)

$$Y \overset{D}{=} \tau Y' + (1-\tau)Y'' + 2\tau \log \tau + 2(1-\tau)\log(1-\tau) + 1$$

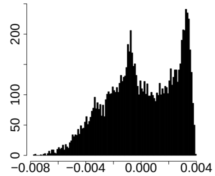
Cophenetic (sum of cophenetic values of tips' pairs)

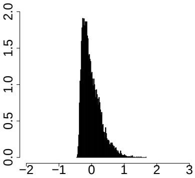
$$Y \overset{D}{=} \tau^2 Y' + (1-\tau)^2 Y'' + \frac{1}{2} - 3\tau(1-\tau)$$

Other Indices (Yule, $\beta = 0$)

Graphics: references on next slide

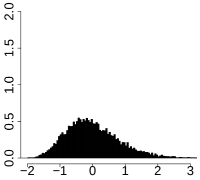
QIB

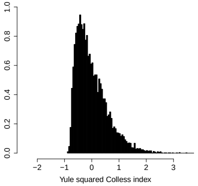


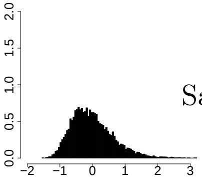


Cophenetic

Colless'







Sackin's

Quadratic Colless'

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